

Evidence-Grounded Verification and Structure-Preserving Repair of Student Science Notes: A Hybrid-Retrieval Knowledge Maps System for English-Medium O/L Science Education

A Curriculum-Grounded, Citation-Enforced Alternative to Generic Note-Taking and Fact-Checking AI

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Component of the AI STEM Ecosystem (STEM Learn LK) · Knowledge Maps Module · English-medium G10–11 Science curriculum

ABSTRACT

Students revising from handwritten or self-written science notes have no reliable way to check those notes against their official curriculum, and existing AI note-taking tools (flashcard/summary generators) and general-purpose fact-checking systems do not close this loop: they either answer questions without verifiable per-claim citations, or they rewrite text wholesale and discard the student's original structure and wording. This paper describes the Knowledge Maps component of the AI STEM Ecosystem, a system that (1) retrieves supporting passages from an official Sri Lankan Grade 10–11 English-medium science curriculum corpus using a hybrid dense-plus-sparse retrieval pipeline with cross-encoder reranking, (2) verifies each factual claim in a student's note against that retrieval with code-enforced evidence citation — an large-language-model verdict is accepted only if it cites an evidence identifier that genuinely exists among the retrieved passages, (3) falls back to a domain-restricted web search only for claims the curriculum corpus cannot decide, (4) reports a statistically gated, calibration-audited confidence score rather than an unvalidated model-reported one, and (5) repairs only the specific incorrect spans of a note while preserving every other character, heading, and list marker exactly as the student wrote it, with every proposed repair re-verified before it is accepted. We situate this design against twelve recent papers and production systems spanning hybrid retrieval, evidence-grounded generation, claim verification, confidence calibration, commercial AI note tools, and prior curriculum-grounded educational chatbots for Sri Lanka, and show that no single existing system combines curriculum-localized grounding, structurally enforced citation, calibrated confidence, and non-destructive repair. This combination constitutes the system's central novelty and addresses a concrete, evidenced research gap.

Keywords: retrieval-augmented generation, hybrid retrieval, reciprocal rank fusion, claim verification, evidence grounding, calibrated confidence, structure-preserving text repair, curriculum-grounded AI, science education, English-medium O/L

1. Introduction

Revision from self-written or handwritten notes is a core study habit for O/L science students, but a note is only useful if it is correct, and a student has no independent, curriculum-grounded way to check it. General-purpose AI assistants can answer a question about a topic, but answering a question and auditing an existing piece of student writing are different problems: the second requires locating every discrete factual claim in the student's own words, deciding whether each is supported by the syllabus, and — critically — fixing only what is wrong without silently rewriting everything else. Consumer AI note-taking products (flashcard and mind-map generators, AI summarizers) address note *creation*, not note *verification*: they compress or restructure whatever text is given to them and do not check it against an authoritative source. General fact-checking and retrieval-augmented generation (RAG) research addresses claim verification, but is rarely built around a fixed, localized, low-resource curriculum, and rarely produces a corrected artifact that preserves the original document's structure.

This paper documents the Knowledge Maps component of a larger AI STEM Ecosystem built for English-medium Sri Lankan Grade 10–11 science students. (Sinhala-medium support exists elsewhere in the wider ecosystem, in a separate Voice Tutor component; it is explicitly out of scope for the system described here, which targets English-medium instruction only.) The component's task is: given a student's note (typed or extracted by OCR from a handwritten page), decide, claim by claim, whether it is correct according to the official syllabus and teacher-guide

corpus; and, where it is not, propose a minimal, evidence-confirmed correction that preserves the student's original structure.

2. Related Work

We group prior work into five threads that each solve one part of this problem.

2.1 Hybrid retrieval and reranking for RAG

Recent work confirms that fusing dense (embedding) and sparse (BM25 lexical) retrieval via Reciprocal Rank Fusion (RRF), followed by a cross-encoder reranker, outperforms either retrieval method alone [2, 3]. Sifei's SemEval-2026 Task 8 system [4] fuses dense and BM25 rankings with RRF and reranks the fused candidates with the BAAI/bge-reranker-v2-m3 cross-encoder — the same reranker model used in the system described here — but is built for open-domain multi-turn dialogue, not for auditing a piece of student writing against a fixed, small curriculum corpus, and includes no claim-level verification, citation enforcement, or confidence calibration stage.

2.2 Evidence-grounded generation and citation

A growing line of work targets citation faithfulness directly: hybrid retrieval-and-reranking frameworks that require generated claims to carry citation labels and verify the cited source supports the claim [1], and structured inline citation generation methods [6]. Crucially, recent analysis of clinical RAG systems identifies *deceptive grounding* — a citation that points to a real, retrieved document, but the wrong entity or fact within it [5] — as a failure mode that survives naive citation-prompting. This is precisely the failure mode this system defends against structurally: an LLM verdict's cited evidence identifiers are validated against the identifiers of the passages that were actually retrieved for that claim before any citation is accepted, and an unmatched or fabricated identifier is discarded rather than trusted (Section 4.2).

2.3 Automated claim / fact verification

Scientific claim verification has its own benchmarks, e.g. SciClaimHunt [7], and surveys of LLM-based automated fact-checking [8] describe the now-standard retrieve–then–verify pipeline. These systems typically stop at a verdict and an explanation; none of the surveyed systems couples the verdict to a structure-preserving rewrite of the original document, and none is built around a fixed school curriculum corpus.

2.4 Confidence calibration for LLM-based systems

Multiple studies show that an LLM's own self-reported confidence is poorly calibrated even after instruction-tuning [9], motivating external calibration methods [10]. Educational tools that surface a confidence signal to a student without any such calibration risk over- or under-trusting the system's verdicts.

2.5 Commercial AI note tools and prior Sri Lankan curriculum-grounded systems

Commercial products (RemNote, Mindgrasp, MarginNote, Evernote AI diagrams, CogniGuide, among others) generate flashcards, summaries, and mind maps from arbitrary input text, optimizing for study-material creation, not correctness auditing against a fixed syllabus. On the curriculum-grounded side, prior Sri Lankan work includes a University of Moratuwa dissertation on a national-curriculum question-answering system [11], SinLlama, a Sinhala-focused base language model [12], and a comparative study of LLMs for mathematics education in Sinhala and Tamil [13]. These systems are answer-oriented (a student asks a question and receives an answer) rather than audit-oriented (a student submits their own note and receives a claim-by-claim, evidence-cited, structure-preserving correction), and none reports a hybrid dense-plus-sparse-plus-reranked retrieval pipeline or a statistically gated confidence layer.

3. Research Gap — The Proof (Paper-by-Paper Comparison)

Table 1 compares the twelve works and system families above against the five capabilities the Knowledge Maps component combines. A checkmark (✓) indicates the capability is present in that work/system as reported in its own description; a dash (—) indicates it is absent or not addressed. No row has a full row of checkmarks — which is the evidence, paper by paper, for the gap this system fills.

Table 1. Paper-by-paper / system-by-system comparison against the five capabilities combined by this work.

WORK / SYSTEM	HYBRID DENSE+SPARSE+ RERANKED RETRIEVAL	CODE-ENFORCED PER-CLAIM CITATION (NOT PROMPT-ONLY)	DOMAIN-RESTRICTED WEB FALLBACK FOR UNDECIDABLE CLAIMS	STATISTICALLY GATED CALIBRATED CONFIDENCE	STRUCTURE-PRESERVING, RE-VERIFIED REPAIR	FIXED LOW-RESOURCE SCHOOL CURRICULUM GROUNDING
Sifei, SemEval-2026 Task 8 [4]	✓	—	—	—	—	—
BM25→Corrective RAG benchmark [3]	✓	—	—	—	—	—
Hybrid Retrieval-Reranking, Evidence-Grounded RAG [1]	✓	partial (prompt-level)	—	—	—	—
Deceptive Grounding analysis [5]	—	identifies the failure, no fix	—	—	—	—
SciClaimHunt benchmark [7]	—	✓ (dataset-level)	—	—	—	—
LLM Fact-Checking survey [8]	varies by surveyed system	varies	varies	—	—	—
Calibration surveys / elicitation studies [9, 10]	—	—	—	✓ (methodology, not deployed)	—	—
Commercial AI note tools (RemNote, Mindgrasp, MarginNote, Evernote AI, CogniGuide)	—	—	—	—	— (rewrites/summarizes)	—
Sri Lankan curriculum QA (UoM dissertation) [11]	—	—	—	—	—	✓
SinLlama [12] / LLMs for Math Ed. in Sinhala&Tamil [13]	—	—	—	—	—	✓

WORK / SYSTEM	HYBRID DENSE+SPARSE+ RERANKED RETRIEVAL	CODE-ENFORCED PER-CLAIM CITATION (NOT PROMPT-ONLY)	DOMAIN-RESTRICTED WEB FALLBACK FOR UNDECIDABLE CLAIMS	STATISTICALLY GATED CALIBRATED CONFIDENCE	STRUCTURE-PRESERVING, RE-VERIFIED REPAIR	FIXED LOW-RESOURCE SCHOOL CURRICULUM GROUNDING
Handwritten Code Recognition for CS Education [14]	—	—	—	—	— (extraction only)	—
This work — Knowledge Maps	✓	✓	✓	✓	✓	✓

Reading across the columns: hybrid dense+sparse+reranked retrieval is well established in general RAG research [1, 3, 4] but is never paired, in the surveyed work, with code-enforced citation. Citation faithfulness is an acknowledged open problem [1, 5, 6], but the proposed fixes stop at prompting or evaluation, not a structural check against the retrieved set. Calibration research demonstrates that LLM confidence is unreliable [9] and proposes calibration techniques [10], but none of the surveyed systems gates a calibrated score behind held-out validation thresholds before ever showing it to an end user, and none is embedded in an education product. No commercial note tool verifies against a curriculum or performs anything narrower than a full rewrite. No prior Sri Lankan curriculum-grounded system [11, 12, 13] combines hybrid retrieval, citation enforcement, calibration, or note repair — they are answer-generation systems, not note-auditing systems. This is the evidenced research gap: *a system that audits a student's own written note, claim by claim, against a fixed low-resource curriculum, with structurally enforced citation, gated calibrated confidence, and a repair step that edits only what is proven wrong.*

4. Proposed System and Novelty

The Knowledge Maps component contributes five design decisions that, individually, echo points made in the literature above, but that no prior system combines:

4.1 Curriculum-localized hybrid retrieval

Passages are retrieved from the official Grade 10–11 English-medium science syllabus and teacher-guide PDFs using dense embedding search (FAISS, cosine similarity over sentence-transformer embeddings) fused by Reciprocal Rank Fusion with BM25 lexical search, then reranked with a BAAI/bge-reranker-v2-m3 cross-encoder. A domain glossary expands queries with syllabus-specific synonyms before retrieval, and every retrieval configuration is fingerprinted into a versioned "retriever signature" so that downstream artifacts (a calibrated confidence model, a scoring policy) can detect and refuse to apply themselves to a retrieval pipeline they were not validated against.

4.2 Structurally enforced evidence citation

The verifying language model is required, via structured tool-calling output, to cite the identifiers of the specific retrieved passages that support its verdict. Unlike prompt-only citation approaches [1, 6], the citation is not trusted at face value: the system independently checks that every cited identifier exists among the passages actually retrieved for that claim, and silently drops any identifier that does not (Section 2.2's "deceptive grounding" failure mode [5] is exactly what this check is designed to prevent). A verdict with no genuinely cited evidence is never reported as decisive.

4.3 Domain-restricted web fallback

Only claims the curriculum corpus cannot decide are escalated to a web search, and only results from an explicit allow-list of approved domains are eligible; a second, independent LLM "judge" step decides the verdict from those

results, and an empty allow-list or a search failure degrades to "insufficient evidence" rather than an unsupported guess.

4.4 Statistically gated calibrated confidence

By default the system reports a transparent, rule-based provisional confidence. An optional logistic-regression calibration model — trained offline on teacher-labelled verdict outcomes — is used only when it satisfies fixed activation gates: at least 100 held-out validation examples, at least 20 high-confidence examples, and a precision at the high-confidence threshold of at least 0.90, with the threshold itself at least 0.75. This is a direct, deployed response to the calibration literature's finding that LLM confidence is unreliable [9]: rather than exposing an uncalibrated or self-reported score, the system either proves a calibration model meets a strict bar or falls back to transparent evidence-strength rules.

4.5 Structure-preserving, re-verified repair

Where commercial note tools rewrite an entire document, this system parses the note into typed blocks (heading, list item, paragraph, blank line) while retaining every original character; each extracted claim is mapped back to its exact character span in the original text via a Unicode-normalization-aware locator that only accepts an unambiguous, unique match; overlapping proposed edits are resolved deterministically; and — critically — a proposed repair is spliced into the note only after a second, independent verification pass confirms the repaired claim is now correct and cited. Unresolved or failed repairs are left exactly as the student wrote them, with the issue surfaced separately rather than silently altering text the system could not confirm.

5. System Architecture Overview

Student note (typed or OCR-extracted) → Claim extraction (structured LLM output) → Hybrid retrieval (dense + BM25 + RRF + reranker) → Per-claim verdict + evidence-ID validation → Domain-restricted web fallback (undecidable claims only) → Gated calibrated confidence → Span-mapped repair proposal → Second-pass re-verification → Structure-preserving corrected note + revision history

Independent per-claim work (verification calls, repair proposals) is executed through a bounded thread-pool fan-out that isolates any single claim's failure from the rest of the batch, so one malformed or slow claim cannot fail an entire note's audit. All curriculum-document citations resolve through a path-traversal-safe, hash-addressed identifier rather than exposing raw file paths, and the module additionally provides handwriting OCR (with a separate region-level confidence review pass) and curriculum-grounded synthesis of flashcards, mind maps, and structured notes from verified content.

6. Conclusion and Future Work

This paper presented the Knowledge Maps component's design and situated it against twelve recent papers and production systems. No surveyed work combines curriculum-localized hybrid retrieval, structurally enforced evidence citation, a domain-restricted web fallback, statistically gated calibrated confidence, and non-destructive, re-verified note repair; the comparison in Table 1 makes this gap explicit, paper by paper. Future work includes a formal user study measuring correction acceptance and learning-outcome impact among English-medium O/L science students, extension of the calibration model as more teacher-labelled data becomes available, and a broader evaluation of the hybrid retriever's recall/precision on the full syllabus corpus against the dense-only and BM25-only baselines it was designed to outperform.

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